

SOHAM NIGAM

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Summary

AI Engineer and CS (AI & DS) student with production experience building autonomous AI agents, RAG pipelines, voice systems, and full-stack ML applications. Owns projects end-to-end from system design to cloud deployment. Seeking AI/ML engineering internships with real-world product impact.

Technical Skills

Languages:	Python, Swift, JavaScript
AI / ML:	NLP, LLMs, RAG, Computer Vision, XGBoost, LightGBM, LSTM, Scikit-learn
Frameworks & Tools:	LangChain, FastAPI, Gemini API, OpenAI API, Pinecone, MongoDB, DVC, MLflow
Infrastructure:	Docker, JWT Auth, REST APIs, CI/CD, AWS S3, Polygon (Web3/Solidity)
iOS:	SwiftUI, MapKit, CoreLocation

Experience

Artificial Intelligence Intern

June 2025 – July 2025

Stark Digital Media Services Pvt. Ltd., Pune

- Built a **GST Invoice Chatbot** using NLP-driven intent detection to guide users through invoice workflows, reducing manual support overhead.
- Engineered a **real-time AI Voice Agent** with speech recognition and NLU pipelines for low-latency conversational user support.
- Architected an **enterprise RAG chatbot** backed by Pinecone for secure, context-aware knowledge retrieval across business documents.
- Developed an **AI document verification system** using Computer Vision and ML to automate validation and eliminate manual processing.

Projects

RoadRupture – AI Pothole Prediction & Safe Route Navigation (iOS)

GitHub

SwiftUI, MapKit, FastAPI, LightGBM, LSTM, Prophet, GeoPandas, CoreLocation

- Native iOS app predicting pothole formation **24–72 hours in advance** using LightGBM (risk classification) and LSTM/Prophet (forecasting) trained on rainfall, elevation, traffic, and road-type features.
- Delivered geospatial risk heatmaps, color-coded route overlays, GPS alerts, and crowdsourced pothole reporting with photo capture to continuously improve model accuracy.

LabChain Sentinel – Autonomous Lab Procurement Agent

GitHub

Python, FastAPI, Gemini API, Pydantic, React

- Built a **closed-loop autonomous AI agent** (Observe → Filter → Reason → Execute) on Gemini that detects stock-out risk and auto-generates purchase orders; Pydantic guardrails ensure 100% UI stability with NABL/ISO 15189-compliant audit logs.

AutoReferral – AI Healthcare Referral System

Live Demo

FastAPI, React, Agentic AI | GDG Build & Grow AI Hackathon 2.0

- Designed and **deployed within 24 hours** an agentic system that scores referral options with calibrated confidence and outputs ranked, explainable healthcare recommendations.

BluCarbon – Blockchain Carbon Credit Platform (Team)

GitHub

React, FastAPI, MongoDB, Solidity, Polygon, Web3.py, Sentinel Hub, IPFS

- Co-built a full-stack blue carbon management platform with satellite NDVI analysis, CNN field-data credibility scoring, and **NFT carbon credits** minted on Polygon via custom Solidity smart contracts.

MLOps Pipeline – End-to-End Production ML System

GitHub

DVC, MLflow, FastAPI, Docker, OAuth2/JWT, Streamlit, CI/CD

- Containerized MLOps lifecycle: DVC versioning, MLflow tracking, hyperparameter tuning, cloud deployment, and a JWT-secured FastAPI inference backend with custom error-handling middleware.

Education

MIT World Peace University, Pune

Aug 2023 – Jun 2029

B.Tech – Computer Science, Artificial Intelligence & Data Science